

DETROIT LAKES, MN, USA

WMO: 727457

Lat: 46.833N

Lon: 95.883W

Elev: 426

StdP: 96.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-26.0	-32.5	0.2	-28.7	-29.0	0.3	-25.1	12.5	-12.8	11.5	-12.3	2.4	330	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	30.9	22.1	28.0	20.2	27.3	19.6	23.5	28.2	22.2	26.7	21.1	25.3	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.8	26.4	21.0	16.5	25.5	19.8	15.2	24.3	72.1	28.6	67.6	26.6	63.1	25.3	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.7	8.6	DB	-32.1	33.6	4.0	1.9	-35.0	34.9	-37.4	36.0	-39.6	37.1	-42.6	38.4	(4)
(5)			WB	-32.3	25.5	4.0	1.8	-35.1	26.8	-37.4	27.8	-39.7	28.8	-42.5	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-12.3	-10.2	-2.7	5.6	12.8	18.4	21.1	19.9	15.3	7.5	-0.9	-8.6	(6)	
	DBStd	13.02	8.16	7.49	7.23	5.64	4.82	3.71	3.11	3.16	4.53	4.96	6.16	7.08	(7)	
	HDD10.0	2854	693	564	395	152	27	1	0	0	8	110	328	577	(8)	
	HDD18.3	4899	951	798	652	382	181	43	11	19	113	338	577	835	(9)	
	CDD10.0	1239	0	0	2	19	113	254	344	308	166	32	2	0	(10)	
	CDD18.3	242	0	0	0	0	9	46	96	69	21	1	0	0	(11)	
	CDH23.3	1817	0	0	0	8	96	351	711	487	153	12	0	0	(12)	
	CDH26.7	416	0	0	0	1	19	77	176	110	32	1	0	0	(13)	
(14) Wind	WSAvg	3.9	4.0	3.9	4.1	4.2	4.2	3.7	3.3	3.3	3.8	4.3	4.2	3.9	(14)	
(15) Precipitation	PrecAvg	664	17	14	27	49	78	112	97	87	77	59	24	18	(15)	
	PrecMax	935	66	52	68	131	224	235	283	201	251	161	99	47	(16)	
	PrecMin	320	2	0	3	2	11	31	10	8	9	2	0	2	(17)	
	PrecStd	138	13	10	16	32	47	52	50	46	49	45	21	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	7.4	17.4	23.9	29.0	32.1	32.5	32.4	30.3	25.0	17.0	6.7	(19)	
		MCWB	2.5	5.1	11.8	13.3	17.2	22.4	23.2	22.6	20.8	16.1	11.4	3.8	(20)	
	2%	DB	2.2	4.1	12.7	20.9	26.1	28.6	30.9	29.1	27.3	21.1	12.6	3.7	(21)	
		MCWB	0.8	2.3	8.3	11.5	16.4	20.6	22.8	20.9	19.4	14.3	8.1	1.7	(22)	
	5%	DB	0.2	2.4	9.7	17.4	23.0	27.1	28.0	27.5	25.0	17.5	10.0	2.1	(23)	
		MCWB	-0.4	1.1	6.5	9.3	14.4	18.9	20.8	20.0	18.1	12.3	6.6	0.8	(24)	
	10%	DB	-1.1	0.2	6.2	15.0	21.2	25.2	27.2	26.2	22.7	15.1	7.3	0.2	(25)	
		MCWB	-1.5	-0.5	3.5	8.0	13.9	17.7	20.2	19.2	17.1	10.4	4.5	-0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	5.4	12.8	14.5	20.1	24.1	25.2	24.6	22.6	17.4	12.1	4.2	(27)	
		MCDB	4.9	7.9	15.0	22.5	25.7	29.4	29.7	29.3	27.3	21.6	15.1	6.4	(28)	
	2%	WB	0.9	2.8	8.9	11.9	17.9	22.1	23.9	22.8	20.8	15.1	8.9	2.1	(29)	
		MCDB	1.9	4.4	12.3	17.7	23.2	27.2	28.6	27.2	24.8	19.7	12.2	3.7	(30)	
	5%	WB	-0.4	1.1	6.2	10.3	16.3	20.5	22.6	21.6	19.5	13.0	6.6	0.7	(31)	
		MCDB	-0.1	2.2	9.2	16.4	21.2	25.0	27.0	25.6	23.6	17.1	9.6	1.5	(32)	
	10%	WB	-1.7	-0.5	3.8	8.7	14.8	19.2	21.4	20.5	18.1	11.1	4.4	-0.4	(33)	
		MCDB	-1.2	0.2	6.2	14.1	19.7	23.5	25.6	24.5	21.6	14.1	7.0	0.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	8.9	10.1	10.5	9.6	9.6	9.8	9.7	8.2	7.1	7.8	(35)	
		MCDBR	8.8	9.6	12.3	14.5	14.0	11.6	10.8	11.3	12.2	12.5	10.9	8.5	(36)	
	5% WB	MCWBR	7.7	7.9	8.1	7.6	6.9	5.7	5.4	5.4	6.1	7.1	7.0	6.9	(37)	
		MCWBR	8.2	8.9	11.9	13.4	12.0	9.9	10.1	10.3	10.4	11.2	10.3	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.276	0.303	0.297	0.350	0.366	0.376	0.378	0.381	0.348	0.306	0.279	0.260	(40)	
		taud	2.303	2.229	2.421	2.353	2.370	2.372	2.376	2.381	2.455	2.554	2.521	2.406	(41)	
	Ebn at Noon		814	858	934	905	898	889	881	864	865	862	824	803	(42)	
		Edh at Noon	75	100	97	115	117	118	116	111	94	73	61	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.62	2.76	3.85	4.87	5.42	5.96	6.31	5.30	3.95	2.44	1.50	1.13	(44)	
	RadStd		0.17	0.28	0.39	0.41	0.37	0.40	0.27	0.44	0.28	0.37	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page